

A15536W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics

Honour School of Economics and Management

Honour School of History and Economics

MICROECONOMIC ANALYSIS

OUTLINE ANSWERS

TRINITY TERM 2019

Thursday 30 May, 09:30–12:30

Please start the answer to each question on a separate page.

There are 6 questions in this paper.

*Answer **four** questions.*

For multipart questions, the weight assigned to each part is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

1. (a) (i) The characteristic polynomial of A is $(2 - \lambda)^3 - 3(2 - \lambda) + 2$. It is simple to verify that the roots of this polynomial are $\lambda = 1, 1, 4$, so these are the eigenvalues.

For $\lambda = 1$, eigenvectors must satisfy

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

so we have $x + y + z = 0$. Therefore, corresponding eigenvectors can be expressed as (non-null) linear combinations of two independent representative eigenvectors, e.g.

$$\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

For $\lambda = 4$, eigenvectors must satisfy

$$\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

and using the Gaussian method, we obtain

$$\left(\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 0 & -3/2 & 3/2 & 0 \\ 0 & 3/2 & -3/2 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 0 & -3/2 & 3/2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

so eigenvectors must satisfy $x = y = z$. We can express eigenvectors as (non-null) multiples of

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

- (ii) The matrix V consisting of two independent eigenvectors associated with the first eigenvalue and one eigenvector associated with the second eigenvalue is invertible.

We then know that

$$V^{-1}AV = D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

Using our eigenvectors,

$$V = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \quad \text{with} \quad V^{-1} = \frac{1}{3} \begin{pmatrix} -1 & 2 & -1 \\ -1 & -1 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

and hence

$$A^6 = VD^6V^{-1} = V \begin{pmatrix} 1^6 & 0 & 0 \\ 0 & 1^6 & 0 \\ 0 & 0 & 4^6 \end{pmatrix} V^{-1} = \begin{pmatrix} \mathbf{1366} & 1365 & 1365 \\ 1365 & \mathbf{1366} & 1365 \\ 1365 & 1365 & \mathbf{1366} \end{pmatrix}$$

- (b) We first prove that whenever the four vectors are a basis of $\mathbf{R}^4$, the intersection of spans consists only of the null vector.

Take w in the intersection of both spans.

So we can write $w = au_1 + bu_2$ and also $w = \alpha v_1 + \beta v_2$.

Then $(au_1 + bu_2) - (\alpha v_1 + \beta v_2) = 0$, or equivalently $au_1 + bu_2 - \alpha v_1 - \beta v_2 = 0$.

The vectors that form the basis have to be independent, so $a = b = -\alpha = -\beta = 0$.

Thus w is the null vector.

We next prove that whenever the intersection of the spans is the null vector, the four vectors form a basis of $\mathbf{R}^4$.

Since there are four vectors, we need only prove that they are independent.

Suppose that $au_1 + bu_2 + cv_1 + dv_2 = 0$; then it follows that $au_1 + bu_2 = -cv_1 - dv_2$.

The vector $w = au_1 + bu_2 = -cv_1 - dv_2$ belongs to both spans, and hence it must be the null vector, i.e. $au_1 + bu_2 = 0$ and $-cv_1 - dv_2 = 0$.

But since $\{u_1, u_2\}$ and $\{v_1, v_2\}$ are each sets of independent vectors, it must be the case that $a = b = 0$ and $-c = -d = 0$,

i.e. $a = b = c = d = 0$, proving that the four vectors are independent.

- (c) (i) Given that $f_x(x, y) = y \frac{x^4 - y^4 + 4x^2y^2}{(x^2 + y^2)^2}$ away from the origin:

$$f_x(0, y) = y \frac{0 - y^4 + 0}{(0 + y^2)^2} = -\frac{y^5}{y^4} = -y.$$

Similarly, $f_y(x, 0) = x \times x^4 / (x^2)^2 = x$.

Formally,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0,$$

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0.$$

$\lim_{y \rightarrow 0} f_x(0, y) = 0 = f_x(0, 0)$ and $\lim_{x \rightarrow 0} f_y(x, 0) = 0 = f_y(0, 0)$,
so the first-order partial derivatives are continuous at the origin.

- (ii) Using the definition of partial derivatives of the functions f_x and f_y , we have

$$f_{xx}(0, 0) = \lim_{h \rightarrow 0} \frac{f_x(h, 0) - f_x(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

$$f_{xy}(0, 0) = \lim_{h \rightarrow 0} \frac{f_x(0, h) - f_x(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{-h - 0}{h} = -1$$

$$f_{yx}(0, 0) = \lim_{h \rightarrow 0} \frac{f_y(h, 0) - f_y(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h - 0}{h} = 1$$

$$f_{yy}(0, 0) = \lim_{h \rightarrow 0} \frac{f_y(0, h) - f_y(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

This gives the Hessian matrix at the origin as in the question.

Since the Hessian matrix is not symmetric, the second-order partial derivatives are not continuous (Young/Schwarz's theorem).

2. (a) When f is (strictly) concave, a local maximum or a stationary point is a (unique) global maximum.

With a concave objective and convex constraints, the Kuhn-Tucker first order conditions are sufficient for a global maximum.

Without differentiability, cannot appeal to Kuhn-Tucker first order conditions.

(b) $f = x^2 + y^2 - cxy$, $Df = (2x - cy \quad 2y - cx)$, $D^2f = \begin{pmatrix} 2 & -c \\ -c & 2 \end{pmatrix}$.

$$\text{Det} = 4 - c^2, \text{Tr} = 4 \dots$$

$0 < c < 2$: $\text{Det} > 0, \text{Tr} > 0 \implies \text{e-vals.} > 0 \implies \text{pos.def.} \implies \text{strictly convex};$

$c = 2$: $\text{Det} = 0, \text{Tr} > 0 \implies \text{e-vals.} = 0, 4 \implies \text{pos.semi-def.} \implies \text{weakly convex};$

$c > 2$: $\text{Det} < 0 \implies \text{e-vals.} +, - \implies \text{indef.} \implies \text{neither}.$

(c) $c = 1$: $f = x^2 + y^2 - xy$, $Df = (2x - y \quad 2y - x)$.

f strictly convex, FOCs nec. & suff. for a global min.

$$2x = y, x = 2y \text{ therefore } x = y = 0; f(0, 0) = 0.$$

Restrict domain to $\{(x, y) : -4 \leq x \leq 4, -4 \leq y \leq 4\}$:

f is still strictly convex on this domain, and the set contains the unique minimising point $(0, 0)$, therefore no other local minima.

Further restrict the domain of f to the boundary of that set.

The boundary is not a convex set, so f is no longer a convex function.

But ...

$$f(4, y) = 16 + y^2 - 4y \text{ is str.convex with min at } y = 2,$$

although f would str.decrease if we moved from boundary to interior;

$$f(-4, y) = 16 + y^2 + 4y \text{ is also str.convex with min at } y = -2,$$

although again f would str.decrease if we moved from boundary to interior;

same for $f(x, 4)$ and $f(x, -4)$, therefore 4 local minima:

at $(4, 2)$, $(-4, -2)$, $(2, 4)$, $(-2, -4)$, and in each case $f = 12$.

- (d) Maximise $f(x, y) = x^2 + y^2 - xy$ on the set $\{(x, y) : -4 \leq x \leq 4, -4 \leq y \leq 4\}$.

Lagrangian $\mathcal{L} = x^2 + y^2 - xy - \lambda[x^2 - 16] - \mu[y^2 - 16]$.

Consider $Dg = \begin{pmatrix} 2x & 0 \\ 0 & 2y \end{pmatrix}$: the matrix of binding constraints is of full rank when one or both constraints bind, so the constraint qualification cannot fail.

Therefore the K-T FOCs are necessary for a max (but not sufficient).

$$\text{FOC}_x : \quad 2x - y - 2\lambda x = 0$$

$$\text{FOC}_y : \quad 2y - x - 2\mu y = 0$$

$$\text{CS}_x : \quad \lambda \geq 0, \quad x^2 \leq 16, \quad \lambda[x^2 - 16] = 0$$

$$\text{CS}_y : \quad \mu \geq 0, \quad y^2 \leq 16, \quad \mu[y^2 - 16] = 0$$

(0) neither constraint binds: K-T FOCs satisfied at $(0, 0)$, but that's a min.

(1) one constraint binds:

$x = +4, -4 < y < 4$, so $\lambda \geq 0, \mu = 0$, and then $\text{FOC}_y \implies y = x/2 = +2$;

$x = -4, -4 < y < 4$, so $\lambda \geq 0, \mu = 0$, and then $\text{FOC}_y \implies y = x/2 = -2$;

in each case, $\lambda = (2x - y)/2x = 3/4$, so candidate local max at $(\pm 4, \pm 2)$.

(Also candidate local max at $(\pm 2, \pm 4)$.)

Either use the argument from part (c) to assert that none of these is a local max, or check SOC using determinant of bordered Hessian:

$$\begin{vmatrix} 0 & 2x & 0 \\ 2x & 2 - 2\lambda & -1 \\ 0 & -1 & 2 - 2\mu \end{vmatrix} = -2x(4x) = -8x^2 < 0, \text{ therefore not a max.}$$

(2) both constraints bind:

$x = +4, y = +4, \lambda = \mu = \frac{1}{2}$ therefore local max at $(+4, +4)$;

$x = -4, y = -4, \lambda = \mu = \frac{1}{2}$ therefore local max at $(-4, -4)$;

$x = +4, y = -4, \lambda = \mu = \frac{3}{2}$ therefore local max at $(+4, -4)$;

$x = -4, y = +4, \lambda = \mu = \frac{3}{2}$ therefore local max at $(-4, +4)$.

3. (a) Standard quasi-linear utility optimization gives

$$x_a = \begin{cases} 0, & m_a p \leq 4 \\ m_a - 4/p, & m_a p > 4 \end{cases} \quad y_a = \begin{cases} m_a/p, & m_a p \leq 4 \\ 4/p^2, & m_a p > 4 \end{cases}$$

$$x_b = \begin{cases} m_b, & m_b \leq p^2 \\ p^2, & m_b > p^2 \end{cases} \quad y_b = \begin{cases} 0, & m_b \leq p^2 \\ (m_b - p^2)/p, & m_b > p^2 \end{cases}$$

(b) For the interior of the Edgeworth box the solution is obtained by equating MRS's:

$$\sqrt{y_a}/2 = 1/\sqrt{x_b} = 1/\sqrt{3 - x_a}, \text{ i.e. } y_a = 4/(3 - x_a).$$

This convex curve intersects the edges of the box at points $(0, 4/3)$ and $(11/5, 5)$.

Thus, the set of Pareto-optimal allocations is:

$$y_a = \begin{cases} [0, 4/3] & \text{when } x_a = 0 \\ 4/(3 - x_a) & \text{when } x_a \in (0, 11/5) \\ 5 & \text{when } x_a \in (11/5, 3] \end{cases}$$

(c) $\omega_a = (3, 3) \implies m_a = 3 + 3p, \quad \omega_b = (0, 2) \implies m_b = 2p.$

Look for an interior solution where $x_a = 3 + 3p - 4/p, \quad x_b = p^2$. Thus,

$$3 + 3p - 4/p + p^2 = 3 \quad \text{leading to } p^* = 1,$$

with equilibrium consumption bundles being

$$(m_a - 4/p, 4/p^2) = (2, 4) \quad \text{and } (p^2, (m_b - p^2)/p) = (1, 1).$$

(d) $MRS^a = 1/2, \quad MRS^b = 1/\sqrt{3}$ so they are not equal on the boundary.

Suppose that the government redistributes endowments to the proposed allocation.

If this is an equilibrium, the demand for good y of consumer a is $y_a = 1$, while that of consumer b is $y_b = (3 + 4p - p^2)/p$.

Market clearing gives $1 + (3 + 4p - p^2)/p = 5 \implies p^* = \sqrt{3}$.

This gives a Walrasian equilibrium.

However, consumer a would like to increase consumption of y , but cannot do so as she cannot decrease consumption of x from 0.

At any $p \in (\sqrt{3}, 2]$, consumer b would demand more than the total endowment of good x , so the price vector is uniquely determined.

4. Initial wealth Y , loss L with probability π , premium $M > \pi L$.

A portion βL can be covered at cost of βM .

(a) Wealth after no loss occurs: $Y - \beta M = Y - M + (1 - \beta)M$

Wealth after loss: $Y - \beta M - L + \beta L = Y - M - (1 - \beta)(L - M)$

Agent's objective:

$$\max_{\beta} (1 - \pi) u(Y - M + (1 - \beta)M) + \pi u(Y - M - (1 - \beta)(L - M))$$

With $w \equiv Y - M$, $\alpha \equiv 1 - \beta$, and $\tilde{x} = M$ w.p. $(1 - \pi)$, $\tilde{x} = -(L - M)$ w.p. π , the maximand becomes $V(\alpha) \equiv E[u(w + \alpha\tilde{x})]$.

(b) $V'(\alpha) = E[\tilde{x} u'(w + \alpha\tilde{x})]$, and the FOC is $V'(\alpha) = 0$.

The SOC for a maximum is $E[\tilde{x}^2 u''(w + \alpha\tilde{x})] < 0$;

this is satisfied when the agent is risk averse, since then $u'' < 0$.

$V'(0) = E[\tilde{x} u'(w)] = u'(w) E[\tilde{x}]$;

note that $E[\tilde{x}] = M - \pi L > 0$ and $u' > 0$, so $V'(0) > 0$.

Given that $u'' < 0$, it must be the case that the FOC is satisfied at a strictly positive value α^* .

(It is possible that $V'(1) > 0$, in which case $\alpha^* = 1$.)

$\alpha^* > 0 \implies \beta^* < 1$.

A risk averse DM should take *some* stake in a favourable gamble.

When insurance is actuarially *unfavourable*, underinsuring is a favourable gamble – so the agent should underinsure.

(c) $\beta^* > 0 \implies \alpha^* < 1$, so the FOC determines α^* : $E[\tilde{x} u'(w + \alpha^*\tilde{x})] = 0$.

Differentiate w.r.t. w :

$$\begin{aligned} E \left[\tilde{x} u''(w + \alpha^*\tilde{x}) \frac{\partial w}{\partial w} + \tilde{x}^2 u''(w + \alpha^*\tilde{x}) \frac{\partial \alpha^*}{\partial w} \right] &= 0 \\ \implies \frac{\partial \alpha^*}{\partial w} &= \frac{-E[\tilde{x} u''(w + \alpha^*\tilde{x})]}{E[\tilde{x}^2 u''(w + \alpha^*\tilde{x})]} \end{aligned}$$

Denominator is unambiguously negative; sign of the numerator is ambiguous.

Assume that absolute risk aversion, $A(y) = -u''(y)/u'(y)$, is decreasing in y .

Since $\alpha^* > 0$, this implies that, for each non-zero realisation x of $\tilde{x}$

$$x A(w + \alpha^*x) < x A(w) \quad \text{so} \quad -x u''(w + \alpha^*x) < x A(w) u'(w + \alpha^*x)$$

Taking expectations (and noting that $\tilde{x}$ has *some* non-zero realisations)

$$-E[\tilde{x} u''(w + \alpha^*\tilde{x})] < A(w) E[\tilde{x} u'(w + \alpha^*\tilde{x})] = 0 \quad \text{by the FOC}$$

LHS is the numerator in the expression for $\frac{\partial \alpha^*}{\partial w}$; hence $\frac{\partial \alpha^*}{\partial w} > 0$ and so $\frac{\partial \beta^*}{\partial w} < 0$.

Experimental and empirical evidence is mostly consistent with decreasing absolute risk aversion.

5. Workers: $u(w, e, \theta) = w - g(e, \theta)$, where $g(e, \theta) = e/\theta$; firm: $\pi(e) = 2e^{1/2}$.

(a) θ public information.

Firm wants to:

$$\max_{(w_L, e_L), (w_H, e_H)} \lambda [\pi(e_H) - w_H] + (1 - \lambda) [\pi(e_L) - w_L]$$

$$\text{s.t.} \quad w_H - g(e_H, \theta_H) \geq 0 \quad \text{and} \quad w_L - g(e_L, \theta_L) \geq 0$$

Let $\alpha \geq 0$, $\beta \geq 0$ be the K-T multipliers on these Participation constraints.

$$\text{FOC}_{w_H}: -\lambda + \alpha = 0, \quad \text{FOC}_{w_L}: -(1 - \lambda) + \beta = 0.$$

Therefore $\alpha = \lambda > 0$ and $\beta = 1 - \lambda > 0$, and hence complementary slackness implies that $w_H = g(e_H, \theta_H)$ and $w_L = g(e_L, \theta_L)$.

$$\text{FOC}_{e_H}: \lambda \pi'(e_H) - \alpha g_e(e_H, \theta_H) = 0, \quad \text{FOC}_{e_L}: (1 - \lambda) \pi'(e_L) - \beta g_e(e_L, \theta_L) = 0.$$

Therefore $\pi'(e_H) = g_e(e_H, \theta_H)$ and $\pi'(e_L) = g_e(e_L, \theta_L)$.

{Informally, the firm chooses effort so that Marginal Benefit = Marginal Cost, and then sets wages to that the participation constraints bind.}

Optimal contracts:

$$\pi'(e_H) = g_e(e_H, \theta_H) \implies e_H^{-1/2} = \theta_H^{-1}, \text{ so } e_H^* = \theta_H^2; \quad \text{similarly } e_L^* = \theta_L^2.$$

$$w_H^* = g(e_H^*, \theta_H) = \theta_H^2 / \theta_H = \theta_H; \quad \text{similarly } w_L^* = \theta_L.$$

$$\text{So } (w_H^*, e_H^*) = (\theta_H, \theta_H^2) \text{ and } (w_L^*, e_L^*) = (\theta_L, \theta_L^2).$$

{In case we need it:

$$\pi(e_H^*) - w_H^* = 2\theta_H - \theta_H = \theta_H, \text{ and } \pi(e_L^*) - w_L^* = \theta_L, \text{ so } E[\text{profit}] = \lambda \theta_H + (1 - \lambda) \theta_L.}$$

(b) θ private information from now on.

(i) The Revelation Principle says that the firm can restrict the form of contracts to ones where the firm asks each worker to reveal his type, and, knowing the effort and wage associated with each possible report, the workers finds its optimal to tell the truth.

Firm wants to:

$$\max_{(w_L, e_L), (w_H, e_H)} \lambda [\pi(e_H) - w_H] + (1 - \lambda) [\pi(e_L) - w_L]$$

$$\text{s.t.} \quad w_H - g(e_H, \theta_H) \geq 0 \tag{1}$$

$$w_L - g(e_L, \theta_L) \geq 0 \tag{2}$$

$$w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H) \tag{3}$$

$$w_L - g(e_L, \theta_L) \geq w_H - g(e_H, \theta_L) \tag{4}$$

(ii) Assume that (3) & (2) are satisfied:

$w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H) \geq w_L - g(e_L, \theta_L) \geq 0$, so (1) is then redundant.

Assume that (3) & (4) are satisfied: reduce w_L (and w_H by the same amount) until (2) binds.

Assume that (4) is satisfied: reduce w_H until (3) binds.

Now (2) gives $w_L = g(e_L, \theta_L)$ and (3) gives $w_H = g(e_H, \theta_H) + g(e_L, \theta_L) - g(e_L, \theta_H)$.

(iii) Temporarily ignore (4).

FOC _{e_H} :

$$\lambda[\pi'(e_H) - \partial w_H / \partial e_H] = 0, \quad \pi'(e_H) = g_e(e_H, \theta_H), \quad \hat{e}_H^{-1/2} = \theta_H^{-1}, \quad \hat{e}_H = \theta_H^2 = e_H^*.$$

FOC _{e_L} :

$$\lambda[-\partial w_H / \partial e_L] + (1 - \lambda)[\pi'(e_L) - \partial w_L / \partial e_L] = 0,$$

$$\lambda[-g_e(e_L, \theta_L) + g_e(e_L, \theta_H)] + (1 - \lambda)[\pi'(e_L) - g_e(e_L, \theta_L)] = 0,$$

$$\pi'(e_L) = g_e(e_L, \theta_L) + \frac{\lambda}{1-\lambda} [g_e(e_L, \theta_L) - g_e(e_L, \theta_H)].$$

$$\hat{e}_L^{-1/2} = \theta_L^{-1} + \frac{\lambda}{1-\lambda} [\theta_L^{-1} - \theta_H^{-1}] > \theta_L^{-1}, \quad \hat{e}_L < \theta_L^2 = e_L^*.$$

$$\hat{w}_H = \theta_H^2 / \theta_H + \hat{e}_L / \theta_L - \hat{e}_L / \theta_H = \theta_H + \hat{e}_L(\theta_L^{-1} - \theta_H^{-1}) > \theta_H = w_H^*.$$

$$\hat{w}_L = \hat{e}_L / \theta_L < e_L^* / \theta_L = w_L^*.$$

(iv) $g(\hat{e}_H, \theta_L) - g(\hat{e}_L, \theta_L) = (\hat{e}_H - \hat{e}_L) / \theta_L$ and $g(\hat{e}_H, \theta_H) - g(\hat{e}_L, \theta_H) = (\hat{e}_H - \hat{e}_L) / \theta_H$.

Since $\hat{e}_H - \hat{e}_L > 0$ and $0 < \theta_L < \theta_H$, we have $(\hat{e}_H - \hat{e}_L) / \theta_L > (\hat{e}_H - \hat{e}_L) / \theta_H$.

6. (a) There is no competitive equilibrium in this economy.

Clearly, if someone sells this security, she has to make a payment in all states.

However, in at least one of the states the consumer's endowment is zero and therefore the payment cannot be made.

- (b) Actually, the necessary number of securities is not three but *two*.

The reason is that the initial allocation in state 1 is Pareto efficient, thus redistribution is required only in two states and this can be achieved using two securities.

- (c) The resulting allocation does not depend on security design, it is the Pareto-efficient one: $x^a = x^b = (1, 1, \frac{1}{2}, \frac{1}{2})$.

It can be achieved with securities $\alpha = (0, 1, 0)$ and $\beta = (0, 0, 1)$.

(Note that a will be selling α and buying β ; vice versa for b .)

Taking logs, consumer a chooses z_α^a, z_β^a to maximise

$$\ln(1 - q_\alpha z_\alpha^a - q_\beta z_\beta^a) + \frac{1}{3} \ln(1 + 0) + \frac{1}{3} \ln(1 + z_\alpha^a) + \frac{1}{3} \ln(0 + z_\beta^a)$$

leading to:

$$\frac{q_\alpha}{1 - q_\alpha z_\alpha^a - q_\beta z_\beta^a} = \frac{1}{3(1 + z_\alpha^a)}, \quad \frac{q_\beta}{1 - q_\alpha z_\alpha^a - q_\beta z_\beta^a} = \frac{1}{3z_\beta^a}$$

which gives

$$z_\alpha^a = \frac{1 - 4q_\alpha}{5q_\alpha}, \quad z_\beta^a = \frac{1 + q_\alpha}{5q_\beta}.$$

Similarly

$$z_\alpha^b = \frac{1 + q_\beta}{5q_\alpha}, \quad z_\beta^b = \frac{1 - 4q_\beta}{5q_\beta}.$$

Solving $z_\alpha^a + z_\alpha^b = 0$ and $z_\beta^a + z_\beta^b = 0$ gives $q_\alpha = q_\beta = \frac{2}{3}$;

so $z_\alpha^a = z_\beta^b = -\frac{1}{2}$ and $z_\beta^a = z_\alpha^b = \frac{1}{2}$.

Actual prices and demands for the securities depend on the specification chosen (but there must be two non-collinear securities paying in states 2 and 3); however, final consumption vectors x^a, x^b are invariant to the choice.